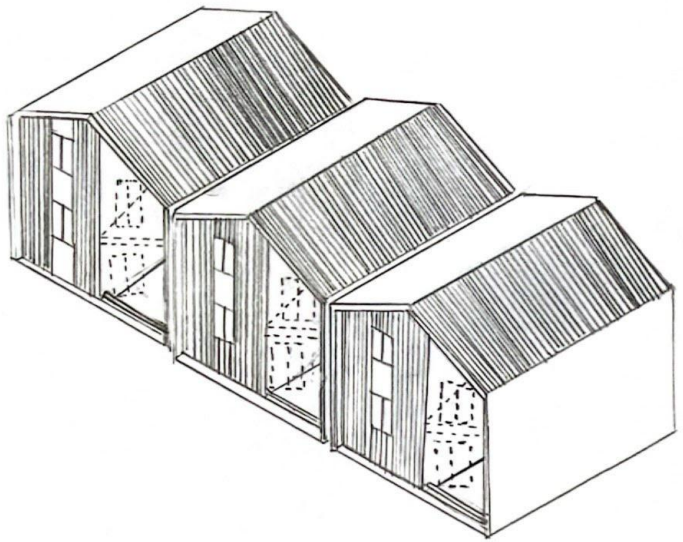
Portfolio Part 1

Villa Verde Community House - designed by Alejandro Aravena

Jarrad Meilak

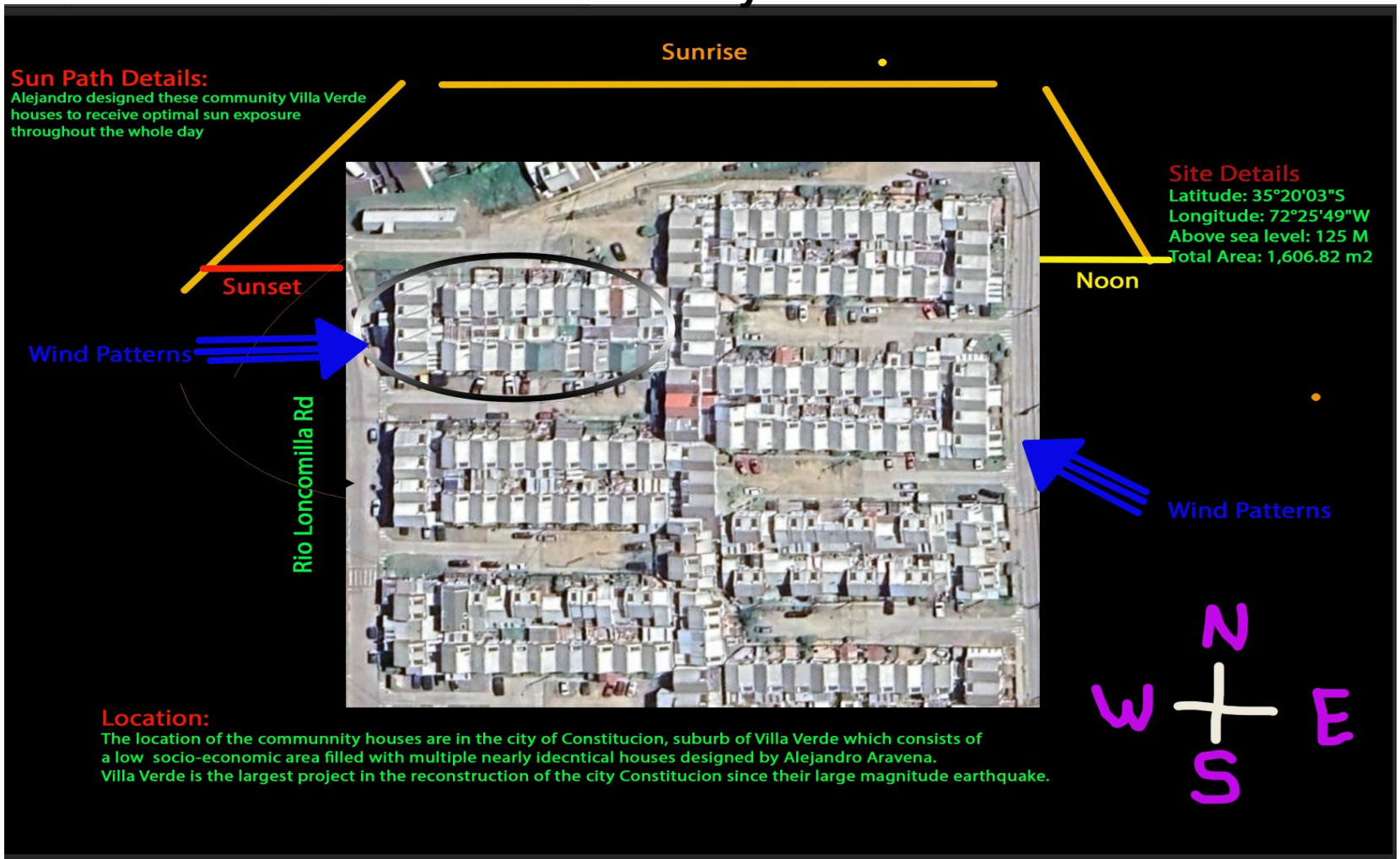


Perspective sketch



Scanned with CamScanner

Site Analysis



Notes/Plan for Model

Materials:

Radiata laser Plywood x 15 pieces of 30cm x 30cm plywood

1x hot glue gun $13.60

3x Hinges $3.00

2x Spray paint, Orange and white $10

1 x Sale ruler

1x T-square

Cost estimation:

Around $40

Measurements:

Each house 30cm x 3 - scale 1:20

Plan/Steps:

- Create to do list for project
- Gather materials for project
- Measure each house and dimensions on base for houses
- Cut house panels to correct size
- Glue panels together
- Glue whole panels to base
- Add minor details



Group Theme - Cost efficiency

Our group 3, Karam, Nathan and myself have chosen the theme of cost efficiency within our selected case studies, as it is an important consideration for our architects to recognise whilst designing their buildings/houses. My chosen architect Alejandro Aravena utilises a wide variety of cost efficiency tactics during the design process of the Villa Verde community houses. His main objective during the design process of the community houses was to support the lower economic class of Chile as they would not be able to afford a large good house, so he's idea was, why not give them half a good house. Constitucion was impacted to a great degree during 2012 when they sustained a large 7.1 magnitude earthquake which wiped out majority of the houses. During the rebuild the government could not afford to rebuild many small houses so they reached out to Alejandro who came up with the great innovative idea of building many half good houses. It was a cheaper and better option during the rebuild process. I chose this case study as I felt it was very interesting and wanted to learn more about this great project of Alejandro's so I thought it would be good to explore further. Alejandro Aravena felt the need to be the person who would help revive his city of Constitucion, Chile after their mass destruction earthquake which destroyed the peoples houses and buildings. He was the person who bought the city back to life.

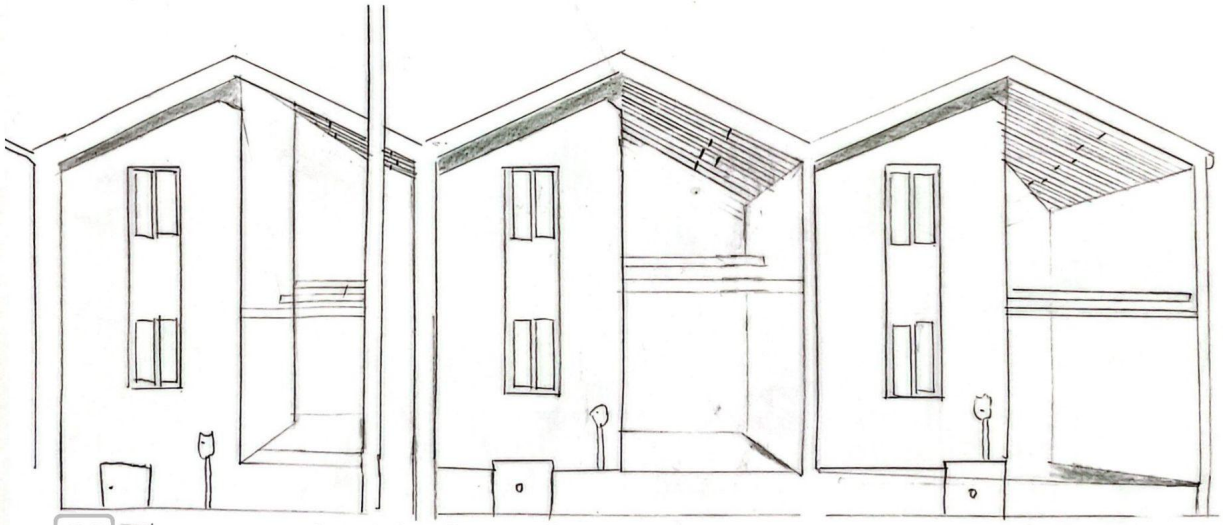


Part 2



Perspective sketch

Front Perspective
Villa Verde Housing

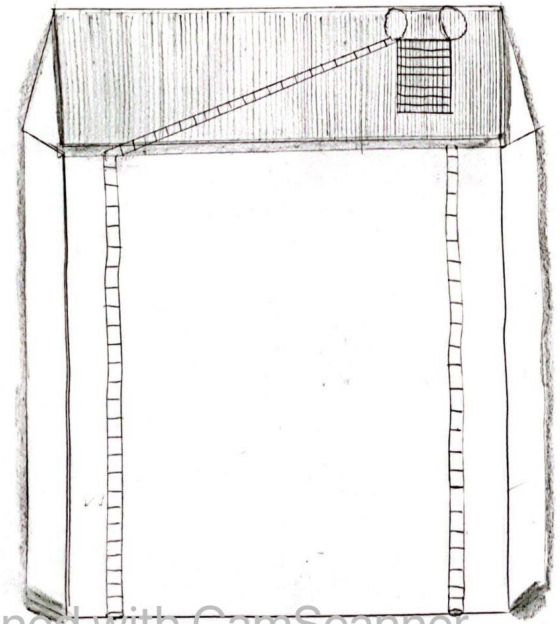


Scanned with CamScanner

Perspective sketch

South East:

Side view Perspective



Scanned with CamScanner

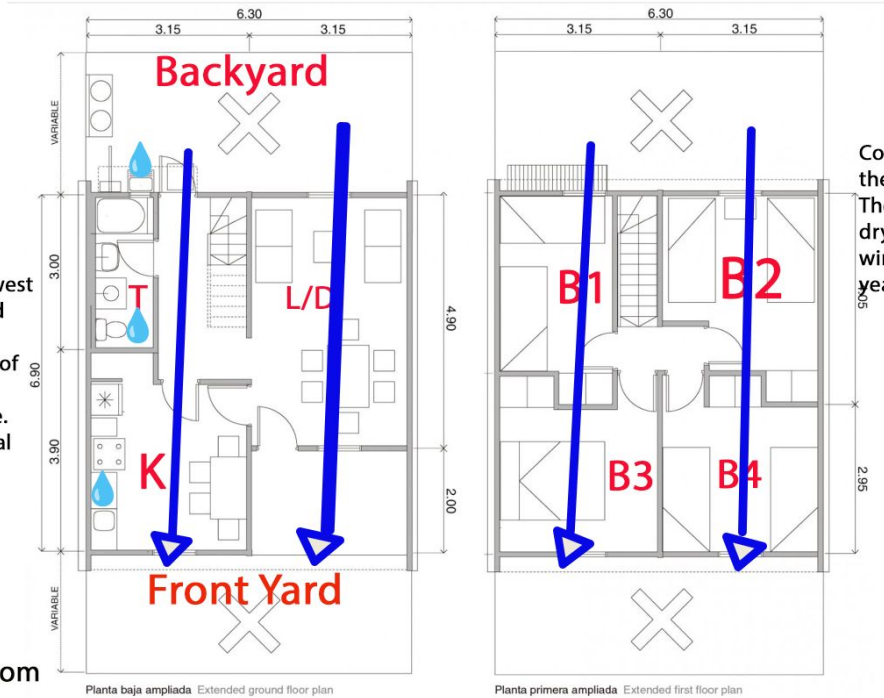
Villa Verde Floor Plan Analysis

Villa Verde Housing

"Half a good house". Alejandro Aravena
Built with the basic necessities to live a normal life.
Once owners have sufficient funds, they can extend the other half of the house to build a full home.

Ventilation

The ventilation in the Villa Verde house, consists of a south - southwest Ventilation direction. The backyard opening allows wind to rush from the backyard through to the front of the house, allowing for natural ventilation in the Villa Verde house. This ventilation would be beneficial during Consiticuion's hot, dry and humid summer days.



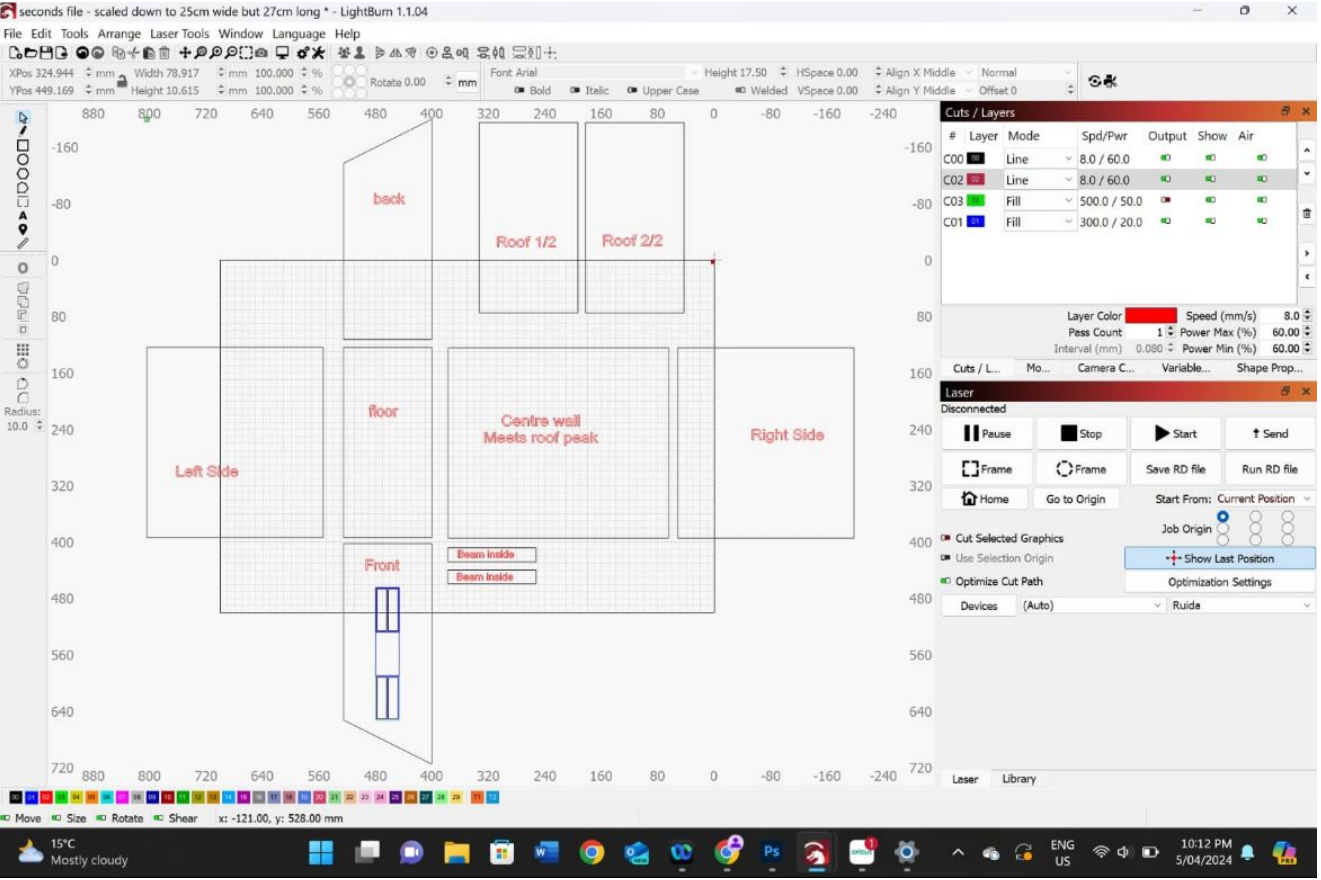
Climate zone Constitucion, Chile

Constitucion, Chile falls under the Mediterranian climate zone. The summer is comfortable and dry, winters are cold and wet & wind is present throughout the year.

Legend

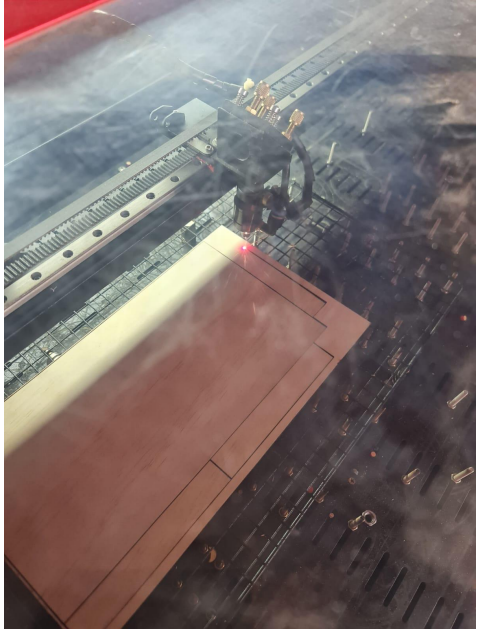
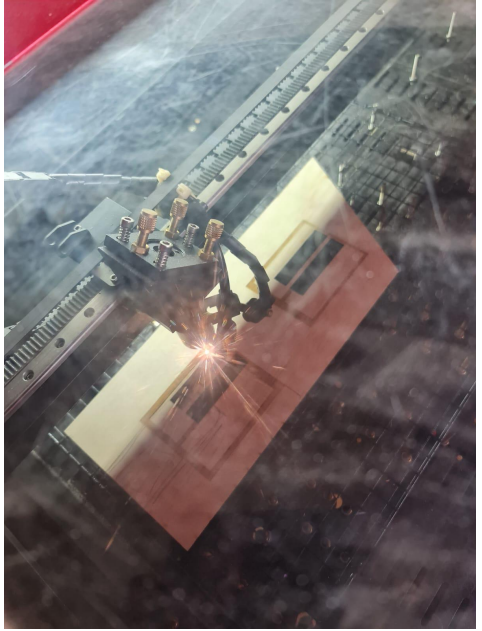
B1, B2, B3, B4 - Bedroom
L/D - Living/Dining
K - Kitchen
T - Toilet
💧 - Water system

Model Making progress



Each piece we will need to construct the Villa Verde house.

The program the laser uses is called lightburn.


Using laser machine to cut out plywood for house.
60W C02 Laser.
Base is 500m x 700mm - not perfectly level and that is important for the cuts and engravings to work correctly. Cutting smaller pieces at a time, rather than putting everything on at once on one big sheet.





Putting pieces of the house together to ensure each piece fits perfectly and there are no flaws.

Group theme and case study

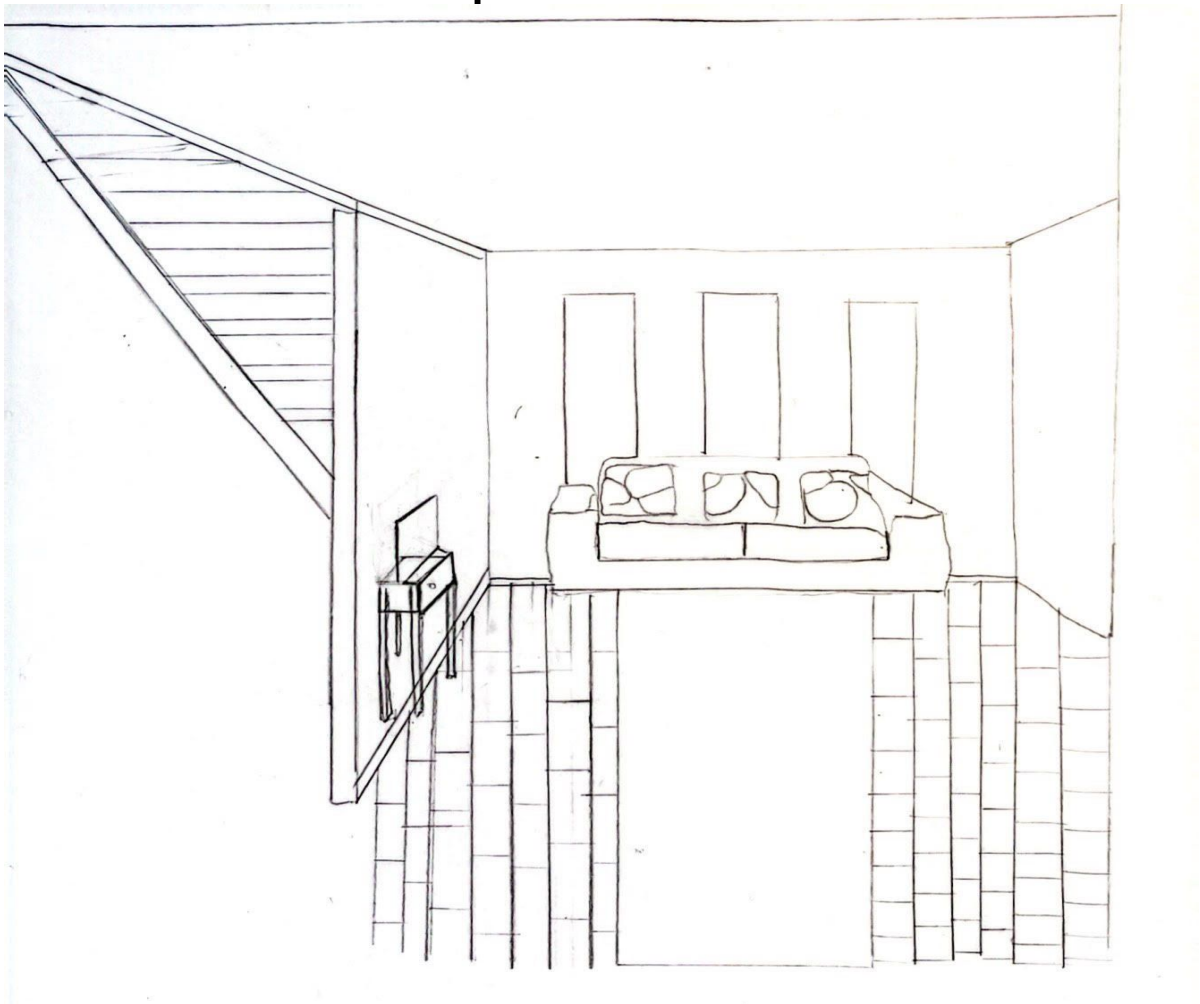
Our group 3, Karam, Nathan and myself have chosen the theme of cost efficiency within our selected case studies, as it is an important consideration for architects to keep in mind when designing a project. My chosen Architect, Alejandro Aravena designed the project of the Villa Verde housing units. Alejandro designed these houses to be social houses, that were constructed after the destructive earthquake that left devastating effects on the local community of Constitution. The Villa Verde houses were designed this way to accommodate a large number of families in the area and provide them with cheap and sustainable housing that is budget friendly with everything they need to live a normal everyday life. They houses were built in half which came at a cheaper price and once the residents had enough money they could extend to make the full house and customise it in any which way they wanted to. There was a lot of leeway with how the residents could customise their own house. It was too expensive to build a full good house so Alejandro thought why not build half a good house. They housing community was designed to fit many locals, many who were affected from the earthquake.

During the designing of my analysis diagram, I learnt how to use the app, photoshop. It was a rough start which took me a couple days to get used to, even know I am still not the best at it, but I believe I have improved from my first analysis diagram. I have learnt about the sun path of Constitution, the wind patterns, side details, location, ventilation and many more interesting facts about the Villa Verde house. Analysing the floor plans helped me to better understand how the locals are living and what appliances/living space they are in.

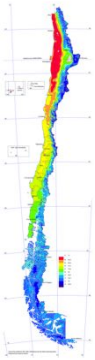
Part 3



Perspective sketch



Villa Verde - Regional Analysis



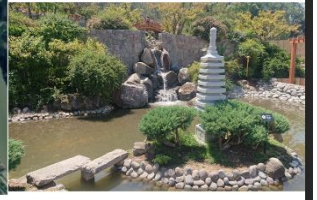
Constitution falls under the mediterranean climate zone.
It has a dry and hot summer and a cool and rainy winter.
Supports a dense population and large agricultural activity.

South Eastern wind from the South Pacific Ocean brings a cool breeze to Villa Verde and Constitution throughout the whole year.









Legend

- Constitution
- Japanese gardens
- Villa Verde
- South Pacific Ocean/ Beach
- Bridge

Topography

Located 800 metres east of Villa Verde is the Japanese Garden landmark which tends to be a spot to attract foreign tourists but also a local hang out spot for the locals of Constitution.
Located 700 metres North of Villa Verde is the south pacific, local beach spot surrounded by a park area, resturants and local hotels which seems to be the busiest location during summer.
Around 1.6km North East of Villa Verde is the heart of Constitucion. 3km East of Villa Verde is the main cross over bridge.

Final Model Photos







Reflection

Throughout the entirety of the case study, Villa Verde Community Housing, designed by Alejandro Aravena, I have developed my knowledge and skills throughout various ways. Initially I was not going to choose this case study but after some deep research about why Alejandro designed the houses the way he did, I was intrigued and interested, therefore finally choosing this case study. I was going to choose the case study of the Australian Islamic Centre designed by Glenn Murcutt, but I felt as if the Villa Verde housing had more of a meaning/story behind it. Assessment stage 1 I found was very difficult in the planning of the model, what resources/materials I would need and how I was going to get started. The next obstacle I came across was learning how to use photoshop. As you can see from the stage 1 site analysis I was not very confident in using this software as I felt it was very difficult. But with persistence and the willingness to learn and become better, I watched plenty of videos and was much more confident in using photoshop for stage 2 then I was in stage 1. I still find it difficult to use but have made an improvement from the first analysis. The first couple of sketches were okay, but what I found most difficult was finding a wider range of photos of the actual house as they were all very similar. I had to use my imagination for the side view sketch which didn't come out too good. I was not able to get a street view on google maps as the site was under construction and being built at that time which made it even harder. Coming to the actual making of the project, was probably my favourite and the part I felt I succeeded in the most. I learnt how to use my cousin's laser cutter machine which took some time to learn, especially the software but eventually the hard work paid off and I have successfully built my version of the Villa Verde Community Housing.

Villa Verde



Designed By
Alejandro Aravena





Villa Verde, Constitution, Chile
Built: 2009-2013



Alejandro Aravena's main objective during the design process of the community houses was to support the lower economic class of Chile. Alejandro designed these houses to be social houses, that were constructed after the destructive 7.1 magnitude earthquake that left devastating effects on the local community of Constitution.

The Villa Verde houses were designed this way to accommodate a large number of families in the area and provide them with cheap and sustainable housing that is also budget friendly with everything they need to live a normal everyday life.



SITE ANALYSIS

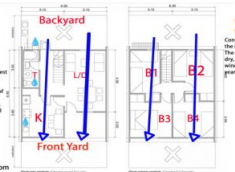
Villa Verde Housing

"Half a good house" Alejandro Aravena
Built with the basic necessities
to live a normal life.
Once owners have sufficient
funds they can extend the
other half of the house to
build a full home.

Ventilation

The ventilation in the Villa Verde house, consists of a south - southwest Ventilation direction. The backyard opening allows wind to rush from the backyard through to the front of the house, allowing for natural ventilation in the Villa Verde house. The ventilation would be beneficial during Constitution's hot, dry and humid summer days.

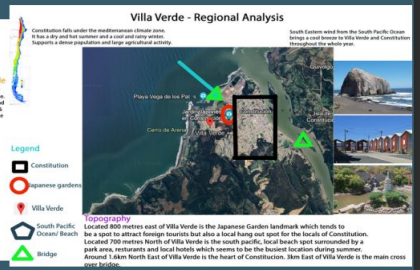
Villa Verde Floor Plan Analysis



Legend
B1, B2, B3, B4 - Bedroom
L/D - Living/Dining
K - Kitchen
T - Toilet
Water system

Climate zone
Constitucion, Chile

Constitucion, Chile falls under the Mediterranian climate zone. The summer is comfortable and dry, winters are cold and wet & wind is present throughout the year.



Topography

Located 800 metres east of Villa Verde is the Japanese Garden landmark which tends to be a spot to attract foreign tourists but also a local hang out spot for the locals of Constitution.
Located 700 metres North of Villa Verde is the south pacific, local beach spot surrounded by a park area, resturants and local hotels which seems to be the busiest location during summer.
Around 1.6km North East of Villa Verde is the heart of Constitucion. 3km East of Villa Verde is the main cross over bridge.

PERSPECTIVE SKETCHES



FINAL MODEL

